



Sprint 2 Presentation



Agenda

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Team Roles & Responsibilities

Charles Varga - DevOps/Tech Architect

Tara Edney - Frontend

Anndeen Daley - Data Engineer/Database

Younes Roumila - Backend

Bella Gerken - AI/NLP Integration

James Simpson - PO/SCRUM Master

QA/QC 1 - Charles Varga

QA/QC 2 - James Simpson



Feedback

- Extensive work on Wiki (from 1am meeting)



Project Description

The Intelligent Academic Path Optimizer is an application which will streamline the academic advising workflow. The app is designed to use AI to facilitate decision making when generating a conflict-free academic schedule for students. By utilizing the Constraint Satisfaction Problem algorithms and predictive modeling, the application will dynamically output the most efficient path to graduation based on specific student parameters.



Team Working Agreement

- Decision Making – Consensus, voting when necessary. Most decisions thus far have been reached by consensus, and the only vote that was taken was for the Project Logo, which was unanimous anyway.
- Team meetings – Decided using When2Meet on a weekly basis. Team has varying schedules week on week, so this is the best as opposed to a set meeting time.
- Team keeps team accountable. We will use group accountability and help remind each other of deadlines.



Personas

Academic Advisor: Isabelle Johnson, 42 · Pace University

Problem

- Manually verifies prereqs and builds schedules for 100+ students
- Changing requirements risk graduation delays

Goals

- Accurate prereq and degree checks
- Automated scheduling
- Earlier risk flags
- More time to help students

How IAPO Helps

- Generates valid pathways so she can focus on mentorship

Graduate Student: William Court, 24 · CS Major

Problem

- Confused about prereqs and availability each semester
- Unclear requirements delay graduation and raise costs

Goals

- Fastest path to graduate
- "What-if" planning before registering
- Balanced schedule
- Steady GPA

How IAPO Helps

- Builds an optimized schedule from his credits, goal date, and load

Dept. Administrator: Michael Rodriguez, 52 · CS Chair

Problem

- Little visibility into which prereqs cause delays
- Relies on manual progress reviews

Goals

- Spot bottlenecks
- Track student progression
- Support data-driven decisions
- Raise grad rates

How IAPO Helps

- Surfaces bottlenecks from degree plans to guide curriculum planning



MVP

- A user-friendly interface in which the user can interact with the AI tied to the app to receive feedback based on the input
- A text input in which students can specify their particular academic needs to the AI
- A clearly visible feedback mechanism which users can utilize to inform developers about the AI, particularly in cases when it generates incorrect information or hallucinates
- A means for users to select the major that they are pursuing and at which level, be it undergraduate or graduate
- Output from the AI which consists of the classes that the students should take based on specified major as well as any text input for specific academic needs in the prompt, and in which semesters these students should take specific classes



Technologies

Frontend (React Website)

- Figma
- React JS
- Native CSS
- Email JS

Backend (Web Server)

- Python 3.12 w. FastAPI
- Gemini API (AI Chat)
- AWS DynamoDB + S3 (Accounts)

Web Hosting

- Github Pages (Frontend)
- PythonAnywhere (Web Server)
- AWS (User Storage)
- Gemini (AI Server)



Algorithms

Input ➡ form fields + free text + optional transcript

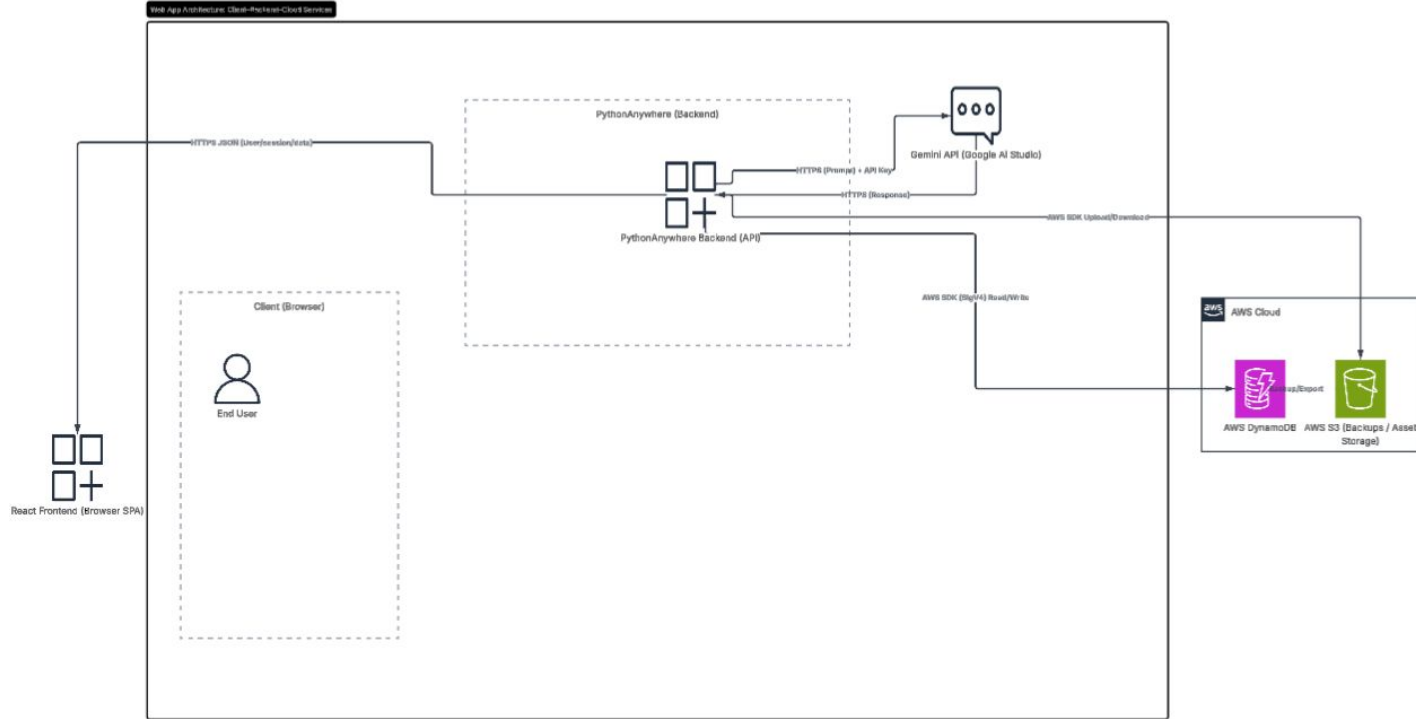
Stage 1: AI Extraction ➡ Receives student inputs, decodes transcript, sends structured prompt to Gemini API, returns validated student profile (completed courses, degree level, constraints, preferences)

Stage 2: AI Catalog Reasoning ➡ Loads full course catalog from database, maps free text to scheduling constraints, scores electives by interest/career fit, checks prerequisite eligibility, calculates remaining degree requirements, drafts candidate plans

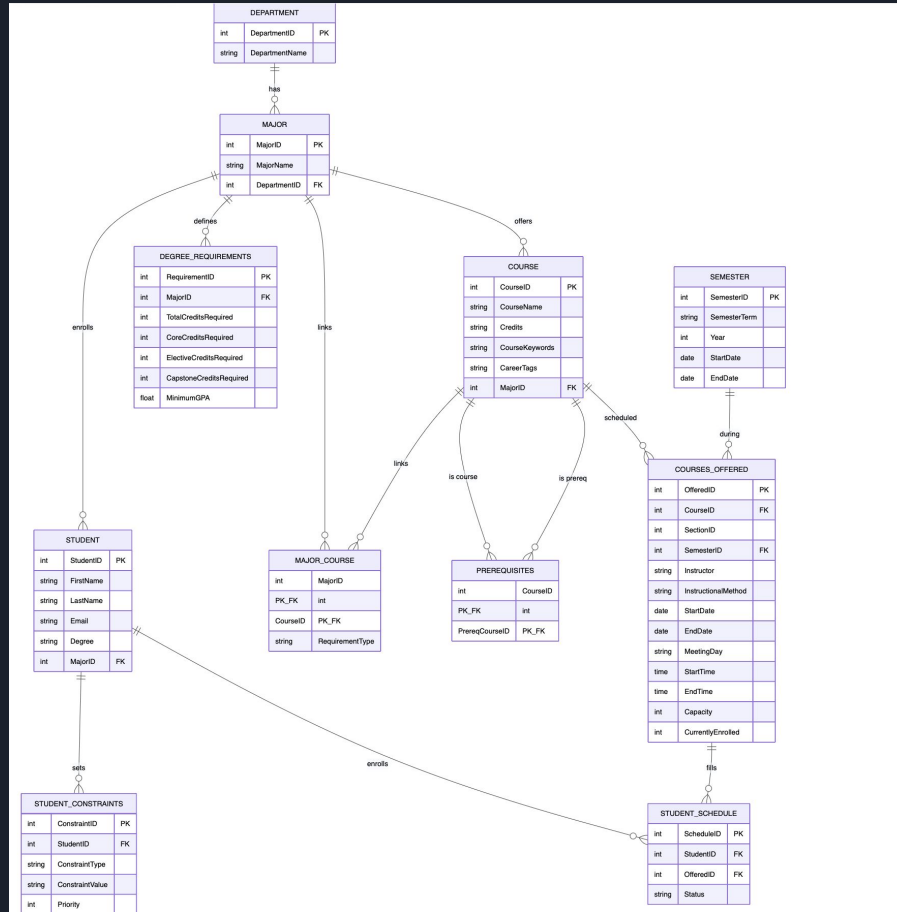
Stage 3: CSP Solver ➡ Validates AI-proposed plans against live section data, resolves conflicts, optimizes final schedule

Output ➡ A personalized, recommended course schedule

Systems Architecture



Database Design





Sprint 1 Recap

- Frontend, Backend, Database and AI were created but not connected.
- Team was still taking care of some planning tasks for Sprint 0 (User stories to get started on coding)

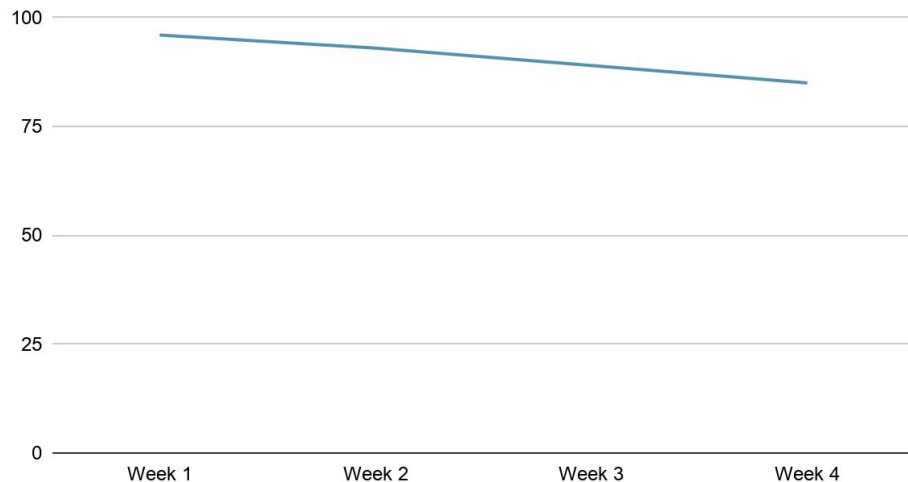


Product & Sprint 2 Backlog

There is no product backlog for Sprint 2. All tickets/cards on the Kanban board were closed. The only case that carried over from Sprint 1 ended up getting scrapped as a poorly-created ticket/user story (James).

Sprint 1 Summary

Burndown

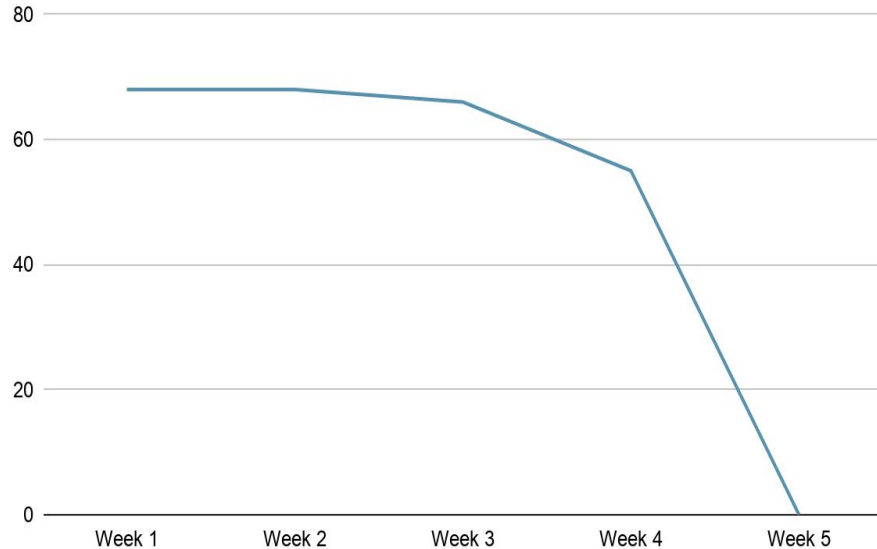


Major Tasks & Stories

- Frontend and Backend were created
- Database and AI provisioned and set up.
- The parts were not connected yet.

Sprint 2 Summary

Burndown

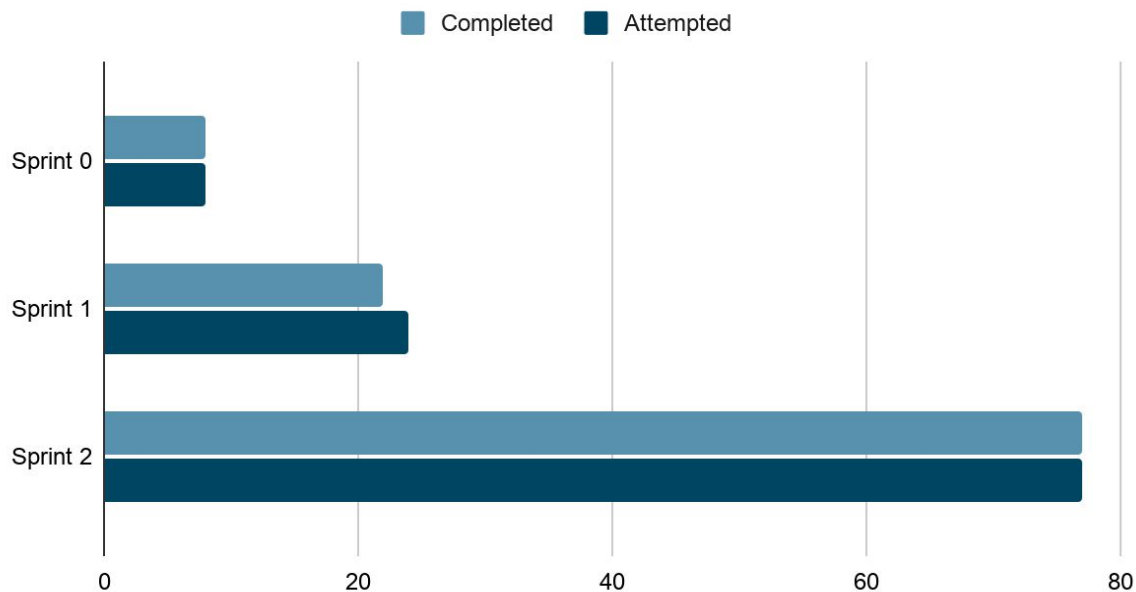


Major Tasks & Stories

- Frontend, Backend, AI and Database connected and working.
- MVP delivered and created (<https://2026-paceu-capstone.github.io/intelligent-academic-path-optimizer>)
- User and Deployment Manuals, Technical Paper and Wiki finished.

Team Velocity

Story Points

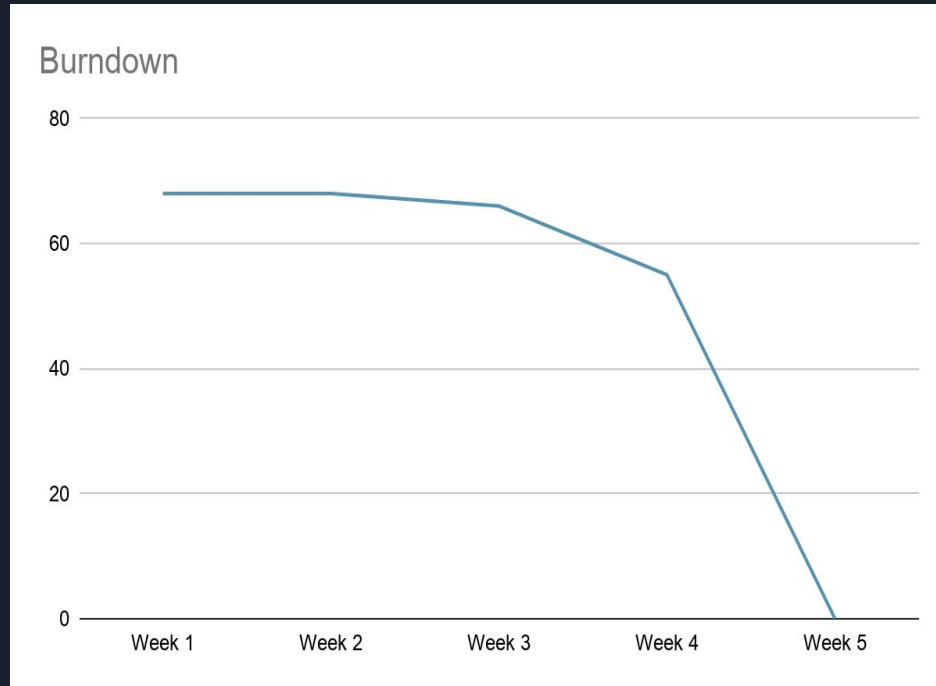


Average Velocity for Sprints 1 & 2:
49.5 Story Points

Completed/Committed Ratios

Sprint 1	22/24 or 91%
Sprint 2	77/77 or 100%
Avg. for both Sprints	95.5%

Sprint 2 Burndown





The Intelligent Academic Path Optimizer

Build the perfect semester schedule with AI. Plan your path to graduation in minutes instead of hours.

[Sign Up](#)

[Log In](#)



AI Schedule Generator

Generate optimized semester schedules based on degree requirements, prerequisites, and your academic goals.



Personalized Planning

Tell the AI your desired graduation semester, preferred credit load, classes you want to take, and any scheduling preferences. We'll build a personalized roadmap.



Upload Your Transcript

Upload your school transcript so IAPO understands the courses you've already completed and generates a more accurate academic plan.

Sprint 2 Demo - Initial Info



Let's Get Started

What's your name?

Degree Level

☐ Undergrad

☒ Graduate

What's your major?

Starting Semester

Ending Semester

How many credits per semester

Next



Upload an optional transcript

5MB Limit



Click to upload your PDF

Back

Next

Sprint 2 Demo - Output





Demo's Schedule

Your Plan

Fall 2026

Spring 2027

Your Plan

SELECTED PLAN

Balanced pace (2 terms, ~7.5 credits/term)

RATIONALE

Schedules your remaining core requirements and best-fit electives across 2 terms at roughly 7.5 credits/term, capstone last.

Completed Courses

No completed courses listed.

Waived / Transferred

None

In Progress

None

Additional Note

All generated schedules are suggestions by AI. There can be mistakes so please double-check results.





Demo's Schedule

Your Plan

Fall 2026

Spring 2027

RECOMMENDED SCHEDULE

Fall 2026 12 Total Credits

Algorithms and Computer Theory

CS608

3 Credits

WHY THIS COURSE

Required core course for your program.

Computer Systems and Concepts

CS604

3 Credits

WHY THIS COURSE

Required core course for your program.

Game Programming

CS617

3 Credits

WHY THIS COURSE

Game Programming — limited overlap with your interests

Intro to Deep Learning

CS672

3 Credits

WHY THIS COURSE

Intro to Deep Learning — limited overlap with your interests

Sprint 2 Demo - AI Output



Major

Computer Science

Starting Semester

Fall 2026

Ending Semester

Spring 2028

Credits

12

Upload Transcript

5MB Limit


Click to upload your PDF

Ask for something else

More math focused, no class on tuesday

Generate Plan

Demo's Schedule

Your Plan

Fall 2026

Spring 2027

Your Plan

SELECTED PLAN

Balanced pace (2 terms, ~7.5 credits/term)

RATIONALE

Schedules your remaining core requirements and best-fit electives across 2 terms at roughly 7.5 credits/term, capstone last.

Completed Courses

No completed courses listed.

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None

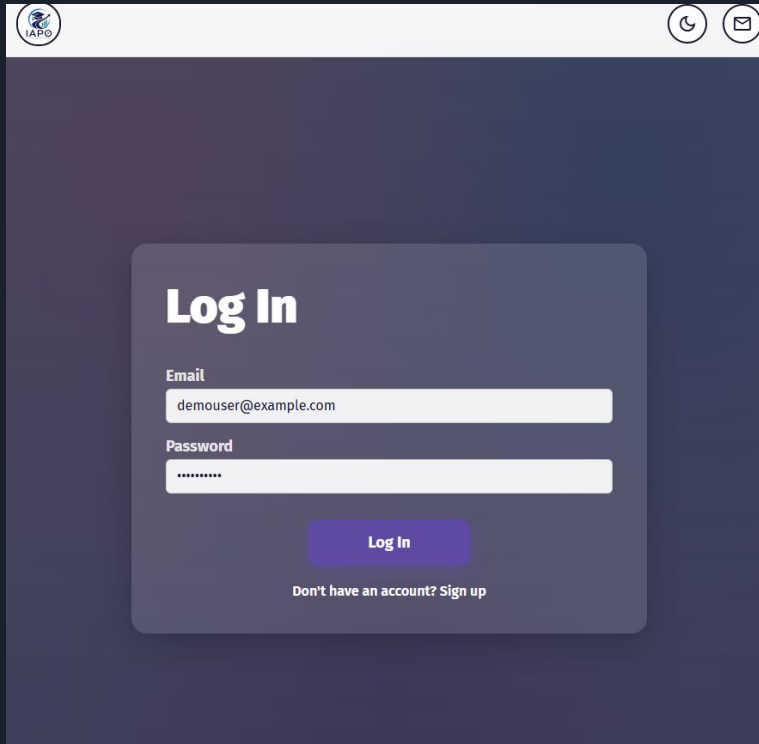
In Progress

None

Additional Note

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Sprint 2 Demo - Log In



The mockup shows a web browser window with a dark blue background. The header bar contains a logo on the left and icons for a moon and an envelope on the right. The main content area features a light blue rounded rectangle with the title 'Log In'. Below the title are two input fields: 'Email' with the text 'demouser@example.com' and 'Password' with masked characters '*****'. A blue 'Log In' button is positioned below the password field. At the bottom of the form, there is a link that says 'Don't have an account? Sign up'.

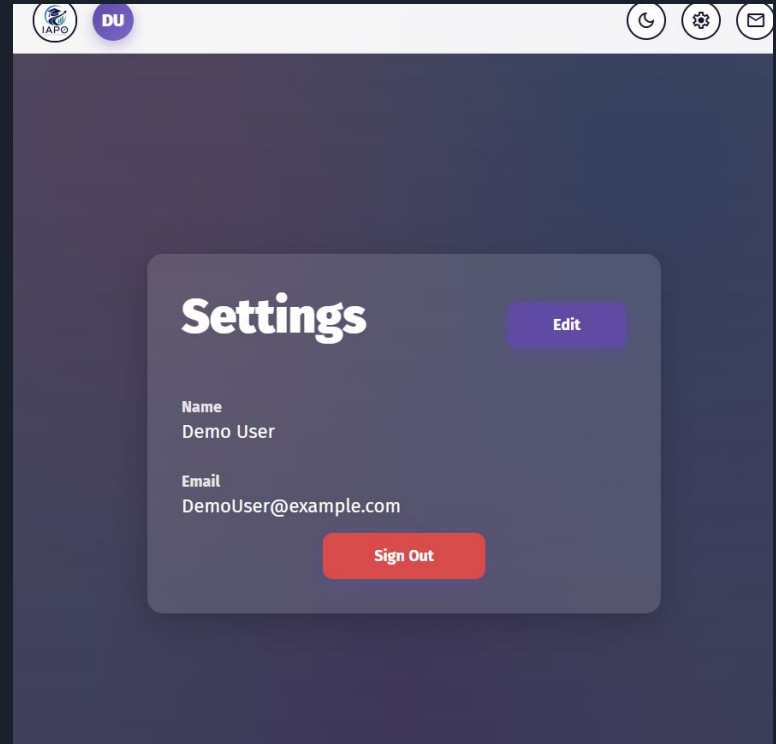
Log In

Email
demouser@example.com

Password

Log In

Don't have an account? Sign up



The mockup shows a web browser window with a dark blue background. The header bar contains a logo on the left, a purple circle with the letters 'DU' in the center, and icons for a moon, a gear, and an envelope on the right. The main content area features a light blue rounded rectangle with the title 'Settings'. In the top right corner of this rectangle is a blue 'Edit' button. Below the title, the user's profile information is displayed: 'Name' with the value 'Demo User' and 'Email' with the value 'DemoUser@example.com'. A red 'Sign Out' button is located at the bottom right of the form.

Settings

Edit

Name
Demo User

Email
DemoUser@example.com

Sign Out

Sprint 2 Demo - Sign Up



Sign Up

Email

test@

Password

Confirm Password

Confirm your password

Please enter a valid email address.

Sign Up

Already have an account? Log in

Sprint 2 Demo - Contact Us





Get in Touch

Have a question, found a bug, or have an idea for a new feature?
We'd love to hear from you.

Name

Email

Message

Tell us how we can help...

Send Message



Sprint 2 Demo - API

Backend:

1. **GET/auth** - Sends a request to confirm if the user has a valid session cookie. This allows them to automatically login on the same browser if they closed the site or returned to it within 7 days.
2. **POST/auth/email** - Used during signup for confirming if a user account already exists, so that the same email can't be used for two different accounts.
3. **POST/signUp** - Used for creating the user account
4. **POST/signIn** - Used for logging into an existing account.
5. **POST/signOut** - Signs the user out and deletes the current session cookies from the user and the database, requiring them to log back in to use IAPO.
6. **POST/updateUser** - Used for updating user credentials (Name, email, password)
7. **POST/updateUser/academic** - Used for updating user academic details (transcript, major, credits per semester, schedule, etc.)
8. **GET/api/info** - Gets the current program info for available majors and semesters

AI:

1. **GET/api/health** - Confirms the service is running, whether the Gemini API key is configured, and if the course catalog is successfully loaded
2. **POST/api/extract** - Accepts the frontend field data and returns a JSON object containing course plan recommendations to be handed to the solver
3. **POST/api/transcriptCheck** - Accepts a PDF that is checked by Gemini to see if it is a valid transcript or not



AI DISCLOSURE - James

What?

- Gemini

How?

- Advice on Agile, assistance with code

Why was this the right tool?

- Gemini provides good code assistance and project organization within the platform.



AI DISCLOSURE - Bella

What?

- Claude

How?

- Generating all code
- Brainstorming for - personas, user stories, acceptance criteria, and test cases. Also to create these documents after I told it the text to include.

Why was this the best solution?

- This was a complex project far past my coding capabilities and I knew Claude could write efficient and accurate code.
- So that I could learn these new concepts and work with the ai to expand on ideas. The document creation was in the interest of saving time and to enhance visualisation.



AI DISCLOSURE - Charles

What?

- Lucidchart, Claude, and Wispr Flow

How?

- Lucidchart was used to generate the systems architecture diagram.
- Claude was used to generate the use case diagram and reorganize the wiki page at the beginning of Sprint 2 into a format that facilitates meeting the checklist requirements.
- Wispr Flow was used as a dictation tool to write out parts of the MVP write-up, the wiki page, the systems architecture section of the technical paper, and my individual report. It converts human voice to text and primarily serves to increase typing speed (triple it in my case), while correcting minor mistakes and removing filler words.

Why was this the best solution?

- Lucidchart had a selection of symbols within its free tier which sufficiently satisfied the requirements to generate the systems architecture diagram via user prompt.
- Claude needed diagram symbols that Lucidchart couldn't generate on its own due to its limited selection of symbols in the free tier. I've also observed it to be the best performing AI for code-related projects, including redesigning wiki pages.
- Wispr Flow outperforms most open source dictation tools which run locally.



AI DISCLOSURE - Anndeen

What?

- [Claude.AI](#), Chat GPT

How?

- Mainly to figuring out how to integrate different tables together to generate results that the backend would understand.
- Thinking of more challenges each persona/users might face so the app could account for more real world use cases.
- To format multiple item entries that can easily be copy and pasted into the database.
- To revise the entity relationship diagram to show the connection between the tables

Why was this the best solution?

- To avoid errors and make sure the data matched and tables worked together correctly for the backend and ai layer
- Helped catch problems that wouldn't be thought on my own
- To save time on repetitive formatting and integration work.



AI DISCLOSURE - Younes

What?

- Claude AI, Chat GPT

How?

- Used for fixing coding errors in the front and backend
- Used for front-end stylization

Why was this the best solution?

- They do a good job of finding the root problem for errors, which is hard to see when viewing several scripts with hundreds of lines of code
- Not well versed with CSS styles and format, so they were used to design the base site, which was then edited to better fit the application.



AI DISCLOSURE - Tara

What?

- Claude

How?

- Aid with clean code

Why was this the best solution?

- As someone new to coding, especially with a team, it's hard to know what the typical best practices are



Links

Github: <https://github.com/htmw/CS691-Summer2026-Team1/wiki>

MVP: <https://2026-paceu-capstone.github.io/intelligent-academic-path-optimizer/>